SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH357

Specimen Examination Paper Solutions

Applied Regression Analysis I

Question 1

(a) $$\hat{\beta}_1 = \frac{\sum (x_j - \bar{x})(y_j - \bar{y})}{\sum (x_j - \bar{x})^2} = \frac{377534.76}{250144.32} = 1.50927$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{25832.05}{88} - 1.50927 \times \frac{16462.34}{88} = 11.204$$

(b) $\hat{\beta}_1 = 1.50927$: it means that an increase (decrease) in house size by 1 square metre is associated with an increase (decrease) in house price of 1.50927 thousands of dollars, or 1509.27 dollars.

(c) $$\hat{\sigma}^2 = s^2 = \frac{SSE}{N-2} = \frac{348053.43}{88-2} = 4047.133$$

(d) $$r^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum \hat{\varepsilon}_j^2}{\sum (y_j - \bar{y})^2} = 1 - \frac{348053.43}{917854.51} = 0.6208$$

<u>Interpretation:</u>

The value of 0.6208 indicates that about 62% of the total sample variation in house prices is explained by the house size.

Question 2

(a) A least-squares estimator for β_1 is a formula in terms of sample data such that the sum of squares of residues SSE is minimized.

$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n}(y_i - \hat{\beta}_1 x_i)^2$$

Setting the first derivative of *SSE* with respect to β_1 equal zero:

$$\frac{d(SSE)}{d\beta_1} = \sum_{i=1}^{n} 2(y_i - \hat{\beta}_1 x_i)(-x_i) = -2\left(\sum_{i=1}^{n} x_i y_i - \hat{\beta}_1 \sum_{i=1}^{n} x_i^2\right) = 0$$

We have

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$

(b) $\sum_{j=1}^{N} \hat{\varepsilon}_j = 0$:The sum of all residuals is equal to zero.

$\sum_{j=1}^{N} x_j \hat{\varepsilon}_j = 0$:The residual and the independent variable x_j are uncorrelated.

(c)(i)

$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$, $\hat{\alpha}_0 = \bar{y} - \hat{\beta}_1 \bar{x}'_j = \bar{y} - \hat{\beta}_1(\bar{x} - \bar{x}) = \bar{y}$

Hence they are not identical.

$$\text{var}(\hat{\beta}_0) = \frac{\sum x_i^2}{n\sum(x_i - \bar{x})^2}\sigma^2, \text{var}(\hat{\alpha}_0) = \frac{\sum x'^2_i}{n\sum(x'_i - \bar{x})^2}\sigma^2 = \frac{\sum x_i^2}{n \cdot \sum x'^2_i}\sigma^2 = \frac{\sigma^2}{n}$$

Hence they are not the same.

(c)(ii)

$$\hat{\beta}_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2},$$

$$\hat{\alpha}_1 = \frac{\sum(x'_i - \bar{x}')(y'_i - \bar{y}')}{\sum(x'_i - \bar{x})^2} = \frac{\sum(x'_i)(y'_i)}{\sum(x'_i)^2} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

Hence they are the same and will have the same variance. (5 marks)

(c)(iii) Model II might be easier to estimate with large x values.

(d) Since

$$\sum x'_j = \sum (x_j - \bar{x}) = 0 \quad \text{and} \quad \sum y'_j = \sum (y_j - \bar{y}) = 0$$

then

$$\bar{x}' = \frac{\sum x'_j}{n} = 0 \quad \text{and} \quad \bar{y}' = \frac{\sum y'_j}{n} = 0$$

Hence

$$\hat{\beta}_0 = \bar{y}' - \hat{\beta}_1 \bar{x}' = 0$$

Therefore, the regression line will pass through the origin.

Question 3

(a) A F-test is required to assess the overall significance of the model:

$H_0: \beta_2 = \beta_3 = 0$
$H_a: \beta_2 \neq 0 \ \cup \beta_3 \neq 0$

The *F* distribution has degrees of freedom v1 = 2 and v2 = 55 - 3 = 52 respectively. The value of the test statistic is given by

$$F_{2,52} = \frac{R^2/p}{(1-R^2)/(n-p-1)} = \frac{0.83/2}{(1-0.83)/52} = 126.94 > 3.1751$$

The model is useful.

(a) Nested model F-test

$$F = \frac{\frac{SSE_{red} - SSE_{Comp}}{1}}{\frac{SSE_{Comp}}{52}} = \frac{\frac{(SST - SSE_{Comp}) - (SST - SSE_{red})}{1}}{\frac{SSE_{Comp}}{52}}$$

$$= \frac{\frac{\frac{(SST - SSE_{Comp})}{SST} - \frac{(SST - SSE_{red})}{SST}}{1}}{\frac{\frac{SSE_{Comp}}{SST}}{52}} = \frac{\frac{0.91 - 0.83}{1}}{\frac{1 - 0.91}{52}} = 46.22 > 4.0266$$

The model is a significant improvement.

(c) The 95% critical *t*-value for a sample size of 30 is 2.052 for 3 parameters and 2.056 for 4 parameters. These levels would suggest that Income (x2) and Age are variables whose parameter estimates are both significantly different from zero but this is not true for the intercept and Price (x3) in both equations. The results indicate that the number of packs purchased increases with Age and Income but Price is not a determinant of the number of packs.

The R^2 value is high in both regressions, indicating that the equations are a good fit. Another point to note is that the coefficient on Income changes substantially between the two regressions when Age is added.

Question 4

(a) Residual vs Predicted Y: The residuals are well behaved between $\hat{Y}$ of around 40 to 120. The model does not appear appropriate as $\hat{Y}$ can reach into negative regions (2 points are < 0) when Y is a biometric figure with minimum value of 0.

The residuals do not appear to be independent of $\hat{Y}$ – the residuals appear to have larger variance at the ends.

Residual vs Creatinine Concentration: The residuals here is very similar to $\hat{Y}$ – almost like a mirror image. X_1 is thus very likely to be a important factor. It has the same problems as $\hat{Y}$, such as non-constant variance. This suggests that the current linear relationship might not be the best fit.

Residual vs Age/Weight: No serious departures and thus no evidence for any lack of fit due to linearity assumption and also no unequal variances.

QQ plot: The points at the upper quantiles deviate significantly from the line, showing that the residuals are not normally distributed.

(b) Yes, there is a need to change.

To normalize the non-normal residuals, a transformation such as the Box-Cox or simpler transformations such as square roots could be used on the Y variable.

Also, the Creatinine Concentration X_1 might require a log transformation or the addition of higher order terms (Quadratic perhaps) as there appears to be a curve in the residuals.

Age and Weight residuals appear more well behaved, although interaction terms with X_1 could be considered.

(a) The theoretical correct answer is :

$$E(\ln(Y)) = \beta_0 + \beta_1(\ln X_1) + \beta_2 \ln(140 - X_2) + \beta_3(ln\, X_3)$$

Able to accept any variants of answers as long as it matches the descriptions in (c).

Question 5

(a) i) Labelling the schools as 1, 2, 3 and 4 is not appropriate. By assigning a number to each school, we are implying that there is a scale or order to the schools.

β_1 in this case will refer to the increase in working hours when you change the x_1 by 1. Changing x_1 by 1 has different meanings. From $x_1 = 1\ to\ x_1 = 2$ means we are comparing Business to Human Development, while $x_1 = 3\ to\ x_1 = 4$ means we are comparing Science and Tech to Others. It also implies that the difference between $x_1 = 1\ and\ x_1 = 3$ (Business and S/T) is double of $x_1 = 2\ and\ x_1 = 3$ (Human Dev and S/T).

ii) No x_1 cannot be set this way. Instead, we should set it this way:

$$x_{1,Biz} = \begin{cases} 1, & \text{if Business School} \\ 0, & \text{if not} \end{cases} \quad ; \quad x_{1,HD} = \begin{cases} 1, & \text{if Human Dev} \\ 0, & \text{if not} \end{cases}$$

$$x_{S/T} = \begin{cases} 1, & \text{if Sci and Tech} \\ 0, & \text{if not} \end{cases}$$

And use this regression equation instead:

$$E(y) = \beta_0 + \beta_{1,Biz}x_{1,Biz} + \beta_{1,HD}x_{1,HD} + \beta_{1,S/T}x_{1,S/T} + \beta_2 x_2 + \beta_3 x_3$$

Under this method, the x_1 is split into 3 binary variables. In this case, $\beta_{1,Biz}$ is the difference in working hours if business school (compared to "others"); $\beta_{1,HD}$ is the difference in working hours if school of human development (compared to "others") and $\beta_{1,S/T}$ is the difference in working hours if Science and Tech (compared to "others").

(b) If $x_1 = 1$,

$$E(y) = \beta_0 + \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_{12}x_2 + \beta_{13}x_3$$
$$= \beta_0 + \beta_1 + x_2(\beta_2 + \beta_{12}) + x_3(\beta_3 + \beta_{13})$$

If $x_1 = 0$,

$$E(y) = \beta_0 + \beta_2 x_2 + \beta_3 x_3$$

This model separates results in the school of S&T against other schools. The interaction term changes both the intercept and slopes for $x_2\ and\ x_3$

----- END OF SOLUTIONS -----